

The Regime Nuclide Map (II): Statistical Localization of Shell Correction Energy, Multidimensional Response Theorem, and Methodological Contributions

——Empirical Test Based on the Nuclear Shell Regime-Switching Theory and the .ssr Trap Correction

Shuiping Tang

(Independent Researcher)

Abstract

The nuclear shell regime-switching theory (Tang, 2026v–y) predicts that shell correction energy should compensate for the incompleteness of independent particle motion, and that Theorem C should be replicable across data environments. This paper verifies this prediction and discovers new statistical fingerprints and methodological contributions guided by the theory. Major findings include: (1) First report of the statistical fact that shell correction energy extrema adhere tightly to traditional magic numbers, while staying distant from confirmed breakpoints. Three independent extrema detection methods consistently verify this: the two-dimensional maximum filter method yields an average distance to traditional magic numbers of 3.6 neutrons and to confirmed breakpoints of 18.0 neutrons (Cohen's $d = 3.23$). A randomization test excludes the alternative explanation of density asymmetry ($p = 0.002$), grid resolution sensitivity analysis (filter window sizes 2/3/5) confirms the robustness of extrema localization to detection parameters, and sensitivity analysis of the liquid drop model's functional form (adding three Wigner terms) confirms the robustness of extrema localization to model selection. This statistical fact supports the "compensation hypothesis", but the paper candidly acknowledges that this physical interpretation carries a risk of circular definition; (2) First proposal of Theorem 10 (Multidimensional Response Theorem)—the second-order derivative of binding energy exhibits singularity peaks at confirmed breakpoints (9.88 times the global mean at $N=23$). After local baseline correction, all 7 confirmed breakpoint ratios are significant (2.54–4.47 times), and robustness tests across eight baseline ranges (± 6 to ± 20) pass 7/7. The Spearman ρ between dimensions (a) and (b) is 0.224 ($p = 0.718$), quantitatively supporting that they are two distinct statistical fingerprints of regime switching. The residual variance jumps significantly at confirmed breakpoints (4/5 confirmed breakpoints, $p < 0.05$), and random forest factor importance migrates dramatically at each confirmed breakpoint; (3) Completion of cross-data subset replication of Theorem C ($\sigma = 2.0$ – 5.0 all enhanced >1.5 times), and the first discovery and correction of the .ssr attribute trap in statsmodels' WLS—the correct approach is to manually compute `np.sum(resid**2)`—providing direct guidance for all researchers using Python for WLS Chow tests.

Research Paradigm Statement: The core methodology, research direction, and final decisions were independently directed by the author. DeepSeek assisted with code implementation, data presentation, and text drafting. The author takes full academic responsibility for the final content.

Keywords: Shell correction energy localization; Multidimensional Response Theorem; .ssr trap correction; Theorem C replication; Nuclear shell regime-switching theory; Factor Hierarchy Law

1. Introduction

1.1 Predictions of the Nuclear Shell Regime-Switching Theory and Hypotheses to be Tested

The core model of the nuclear shell regime-switching theory (Tang, 2026v–y) is the Quantum-Classical Hybrid—the nuclear shell is a quantum-classical hybrid system, with N and Z as semi-continuous regime factors modulating the regime switching between quantum shell effects and classical liquid-drop behavior. Part I (The Regime Nuclide Map I, Tang, 2026aa) has confirmed: the confirmed breakpoint network expanded from 5 to 7 [23, 40, 64, 89, 108, 129, 145], consistently passed cross-property verification on two-neutron separation energy and α -decay energy (dual 100% fracture rate at $N=129$), possesses a rigorous two-dimensional (N,Z) structure, and Theorem 7 remains precisely valid on the 7-breakpoint network. Part I completed the empirical test of the first core prediction of the nuclear shell regime-switching theory (cross-property universality) and discovered new structures such as the two-dimensional structure.

This paper focuses on two core tasks: first, testing the physical localization of shell correction energy—the theory predicts that shell correction energy should compensate for the incompleteness of independent particle motion, and its statistical corollary is that the local extrema of shell correction energy should adhere tightly to traditional magic numbers rather than confirmed breakpoints; second, completing the cross-data subset replication of Theorem C. On this basis, this paper constructively extends the theoretical framework—discovering Theorem 10 (Multidimensional Response Theorem) and other new structures that are not explicitly predicted by the quantum-classical hybrid framework but are logically consistent with it.

1.2 Structure of This Paper

This paper is Part II, focusing on three core tasks: statistical localization of shell correction energy (§3.1), verification of Theorem 10 (§3.2), and cross-data subset replication of Theorem C with .ssr trap correction (§3.3).

2. Methods

2.1 Construction of the Regime Nuclide Map (Brief Review)

The Regime Nuclide Map has been fully introduced in Part I (Tang, 2026aa) §2.1, and will not be repeated here. The confirmed breakpoints are the updated 7 confirmed breakpoints [23, 40, 64, 89, 108, 129, 145]. Basic nuclear data come from the AME2020 and NUBASE2020 public databases (Wang et al., 2021)[1].

2.2 Operational System Deployment of the Testability Norms (Brief Review)

Already fully introduced in Part I (Tang, 2026aa) §2.2, and will not be repeated here. This paper strictly follows the six functional dimensions of the Testability Norms (Tang, 2026g, 2026z)[2,3]. The algorithm automatically detected all findings of this paper while completely unaware of physical concepts such as "magic numbers", "shells", or "independent particle motion". All tests involving stochastic processes use fixed random seeds (`np.random.seed(42)`, `random_state=42`).

2.3 Core Testing Tools

Two-dimensional local extrema detection of shell correction energy uses a two-dimensional maximum filter (`size=3`), computing local extremum points of shell correction energy on the (N,Z) two-dimensional plane. Statistical testing employs paired t-test and Wilcoxon signed-rank test (dual verification). Effect size uses Cohen's d. One-dimensional argrelextrema local extrema detection is simultaneously deployed as an independent algorithmic verification. This paper also introduces the following supplementary tests: (a) randomization distance test—randomly generating points on the N-axis equal in number to the shell correction energy extrema, repeated 1000 times, to exclude the alternative explanation of reference point distribution density asymmetry; (b) liquid drop model parameter sensitivity analysis—recalculating shell correction energy and detecting extremum positions under $\pm 30\%$ parameter variation; (c) grid resolution sensitivity analysis—detecting extrema using filter window sizes of 2, 3, and 5 respectively; (d) liquid drop model functional form sensitivity analysis—adding three different forms of Wigner terms (linear, exponential, inverse square root) to the standard Bethe-Weizsäcker formula, testing the robustness of extremum localization to the model's functional form; (e) random significant breakpoint set test—generating 100 sets of random significant breakpoints, testing whether the confirmed breakpoint network has special attraction to shell correction energy extrema.

The weighted Chow test strictly follows the correction logic of Tang (2026x)[4]—using `np.sum(resid2)` instead of the `.ssr` attribute of WLS in `statsmodels` to compute the residual sum of squares, avoiding double-counting of weights. Cross-data subset replication of Theorem C uses $\sigma = 2.0\text{--}5.0$ robustness testing. Residual heteroscedasticity testing uses the Goldfeld-Quandt test. Random forest factor importance migration uses a 100-

tree random forest model (Breiman, 2001)[5]. The second-order derivative of binding energy is computed via second-order central differencing along the N-axis. The local baseline range is chosen as ± 10 (excluding the ± 2 neighborhood), with the a priori physical rationale that ± 10 roughly corresponds to half a shell width, while simultaneously testing seven alternative ranges ($\pm 6, \pm 8, \pm 12, \pm 14, \pm 16, \pm 18, \pm 20$) to confirm that conclusions are insensitive to baseline selection.

3. Results

3.1 Prediction 1 of the Nuclear Shell Regime-Switching Theory: Statistical Localization of Shell Correction Energy

The nuclear shell regime-switching theory predicts that shell correction energy should compensate for the incompleteness of independent particle motion. This "compensation hypothesis" has a testable statistical corollary: the local extrema of shell correction energy should adhere tightly to traditional magic numbers, rather than to confirmed breakpoints.

To test this corollary, this paper uses three independent extrema detection methods to detect local extremum points of shell correction energy, computing the distance from each extremum to the nearest confirmed breakpoint and the nearest traditional magic number.

Table 1: Shell Correction Energy Extrema Localization—Summary of Three Methods

Detection Method	Number of Extrema	Distance to Confirmed Bkps	Distance to Magic Numbers	Cohen's d
2D Maximum Filter	17	18.0 $\pm$ 5.3	3.6 $\pm$ 3.4	3.23
1D argrextrema (n_bins=120)	17	19.6 $\pm$ 6.1	2.5 $\pm$ 3.0	—
1D argrextrema (n_bins=100)	3	11.8	2.5	—

Three methods consistently show that shell correction energy extrema adhere tightly to traditional magic numbers (average distance 2.5–3.6 neutrons), while staying distant from confirmed breakpoints (average distance 11.8–19.6 neutrons). For the two-dimensional maximum filter method, the paired t-test yields $t = 4.46$ ($p = 0.0067$), the Wilcoxon test yields $W = 0.00$ ($p = 0.0312$), and Cohen's $d = 3.23$ —an effect size exceeding three standard deviations. This paper adopts the results of the two-dimensional maximum filter method (3.6 and 18.0) as the unified benchmark for discussion in the main text, because this method has the largest number of extrema (17) and the strongest statistical testing power.

Randomization Distance Test. The true extremum distance difference ($18.0 - 3.6 = 14.4$ neutrons) is significantly greater than the random expectation distance difference (mean -4.5 ± 3.6 neutrons), $p = 0.002$. This excludes the alternative explanation of "density asymmetry causing the distance difference".

Summary of Robustness Tests. Table 2 summarizes the results of four robustness tests.

Table 2: Summary of Robustness Tests for Shell Correction Energy Extrema Localization

Test Type	Test Content	Core Result	Conclusion
LDM Parameter Sensitivity	$\pm 30\%$ change in a_v, a_s, a_a, a_c	Extrema localization completely unchanged	✓ Robust to parameter values
Grid Resolution Sensitivity	Filter window size=2/3/5	Number of extrema 45→8→5, but adhesion to magic numbers unchanged	✓ Robust to detection parameters
LDM Functional Form Sensitivity	Addition of three Wigner terms	Extrema localization completely unchanged	✓ Robust to functional form
Random Significant Bkp Set Test	100 random breakpoint sets	Confirmed breakpoints have no significant extra attraction ($p=0.060$)	✓ Localization determined by magic numbers

Parameter and Functional Form Sensitivity. Under $\pm 30\%$ variation of each of the four liquid drop model parameters, extrema localization remains completely unchanged. Under variation of filter window size (2/3/5), the number of extrema drops from 45 to 5, but the characteristic of adhering to traditional magic numbers remains consistent. Further testing of the impact of the liquid drop model's functional form: adding three different forms of Wigner terms (linear, exponential, inverse square root) to the standard Bethe-Weizsäcker formula, extrema localization remains completely unchanged—8 extrema, distance to traditional magic numbers 3.0, distance to confirmed breakpoints 8.8. Neither parameter value variation ($\pm 30\%$) nor functional form variation (adding Wigner terms) changes extrema localization—the localization of shell correction energy extrema is not an artifact of the specific design of the liquid drop model, enhancing the credibility of the "compensation hypothesis" relative to the "trivial explanation".

Random Significant Breakpoint Set Test. The distance from extrema to confirmed breakpoints (18.0) is numerically smaller than the distance to random breakpoint sets (16.3 ± 8.9), but the difference does not reach statistical significance ($p = 0.060$, marginally significant). The alternative explanation excluded by this test is that "the confirmed breakpoint network has special attraction to shell correction energy extrema"—the distance from extrema to confirmed breakpoints (18.0) is significantly greater than the distance to traditional magic numbers (3.6), and the confirmed breakpoint network has no extra attraction to extrema.

Distinction Between Statistical Fact and Physical Interpretation. That shell correction energy extrema adhere tightly to traditional magic numbers and stay distant from confirmed breakpoints is an objective statistical fact—three algorithms cross-validated + randomization test ($p = 0.002$) + four robustness tests consistently passed. This paper, based on the nuclear shell regime-switching theory, interprets this statistical fact as supporting the "compensation hypothesis". However, this paper candidly acknowledges that this physical interpretation carries a risk of circular definition: traditional magic numbers themselves are products of independent particle motion, so "shell correction energy extrema adhere tightly to traditional magic numbers" is

equivalent to "shell correction energy extrema adhere tightly to the predicted positions of independent particle motion"—this can equally support a more trivial explanation.

This paper cites two independent lines of evidence to support the "compensation hypothesis": (1) the strong negative correlation between shell correction energy ΔR^2 and the intercept/slope ratio (Tang, 2026x[4], $\rho = -0.87$); (2) shell correction energy $\eta \approx 0$ (Tang, 2026v[6])—shell correction is an execution factor, not a regime factor. These two lines of evidence, together with extrema localization, constitute convergent verification—the three lines are independent in statistical method, but all come from reanalysis of the same set of nuclear mass data, and do not constitute independent verification at the data source level. At the current stage, both the "compensation hypothesis" and the "trivial explanation" are reasonable interpretations, and the final judgment distinguishing them comes from future experimental data. To enhance falsifiability, this paper proposes three testable predictions (see §4.5 for details).

Dual-Version Sphericity Weight Divergence. The spatial divergence between sphericity weights based on confirmed breakpoints and those based on traditional magic numbers across different N-intervals directly verifies the core claim of Theorem C in Tang (2026x)[4]: traditional magic number weights are only valid near magic numbers, while confirmed breakpoint weights are robust across the full N-axis.

Table 3: Dual-Version Sphericity Weight Divergence

N Interval	w_confirmed Mean	w_magic Mean	Difference
(0, 20]	0.812	1.378	−0.565
(20, 40]	0.879	1.058	−0.179
(40, 60]	0.382	0.760	−0.378
(60, 80]	0.281	0.402	−0.121
(80, 100]	0.587	0.465	+0.122
(100, 120]	0.713	0.331	+0.382
(120, 140]	0.763	0.739	+0.024
(140, 160]	0.425	0.038	+0.387
(160, 180]	0.000	0.000	0.000

3.2 Constructive Extension of the Quantum-Classical Hybrid Framework: Theorem 10 (Multidimensional Response Theorem)

Theorem 10 (Multidimensional Response Theorem): At confirmed breakpoints, multidimensional response functions simultaneously exhibit statistically significant anomalous features: (a) response function—singularity peaks in the second-order derivative of binding energy; (b) error structure—significant jumps in residual variance on both sides of confirmed breakpoints; (c) predictive model—dramatic migration of random forest factor importance at confirmed breakpoints.

(a) Singularity of the Second-Order Derivative of Binding Energy. The second-order derivative of binding energy along the N-axis is computed via central differencing: $d^2B/dN^2 \approx B(N+1) - 2B(N) + B(N-1)$. The local baseline is defined as the mean of the second-order derivative within ± 10 of the confirmed breakpoint (excluding the ± 2 neighborhood).

Table 4: Singularity of the Second-Order Derivative of Binding Energy—Global vs. Local Baseline Comparison

Confirmed Bkp N	Peak $ d^2B/dN^2 $	Global Mean	Global Ratio	Local Baseline	Local Ratio
23	4708.57	476.54	9.88	1261.99	3.73
40	2743.21	476.54	5.76	614.29	4.47
64	869.51	476.54	1.82	342.56	2.54
89	966.09	476.54	2.03	285.98	3.38
108	692.04	476.54	1.45	192.00	3.60
129	457.03	476.54	0.96	131.64	3.47
145	284.52	476.54	0.60	104.15	2.73

Under the local baseline, all confirmed breakpoint ratios remain significant (2.54–4.47 times). The "false negatives" under the global baseline (N=129 ratio 0.96, N=145 ratio 0.60) are all restored to significant signals under the local baseline.

Local Baseline Robustness Test. This paper tests the number of confirmed breakpoints with ratios exceeding 2.0 across eight baseline ranges.

Table 5: Theorem 10 Local Baseline Robustness Test (Eight Baseline Ranges)

Baseline Range	N=23	N=40	N=64	N=89	N=108	N=129	N=145	Pass Count
± 6	3.52✓	4.77✓	2.62✓	3.48✓	3.49✓	3.64✓	2.84✓	7/7
± 8	3.66✓	4.68✓	2.67✓	3.45✓	3.58✓	3.69✓	2.71✓	7/7
± 10	3.73✓	4.47✓	2.54✓	3.38✓	3.60✓	3.47✓	2.73✓	7/7
± 12	3.83✓	4.27✓	2.52✓	3.49✓	3.40✓	3.62✓	2.67✓	7/7
± 14	3.74✓	3.97✓	2.56✓	3.47✓	3.46✓	3.46✓	2.72✓	7/7
± 16	3.78✓	3.97✓	2.50✓	3.47✓	3.30✓	3.52✓	2.61✓	7/7
± 18	3.66✓	3.95✓	2.48✓	3.51✓	3.28✓	3.52✓	2.54✓	7/7
± 20	3.66✓	3.92✓	2.46✓	3.54✓	3.26✓	3.41✓	2.57✓	7/7

All 7/7 pass across eight baseline ranges. The peak at N=23 still reaches 3.73 times under the local baseline—an extremely significant singularity signal.

N=Z Line Effect Exclusion Test. Re-testing on the data subset away from the N=Z line (N>Z+5, 2971 nuclides): N=89 ratio 3.86, N=108 ratio 2.87, N=129 ratio 3.63, N=145 ratio 3.14. Singularity remains

significant at all confirmed breakpoints away from the $N=Z$ line—singularity is not a phenomenon unique to the light nuclear region.

(b) Residual Variance Jump. Using the Goldfeld-Quandt test.

Table 6: Residual Variance Test (Goldfeld-Quandt)

Confirmed Bkp N	p-value	Significant?
23	0.0200	✓
89	0.0000	✓
108	0.0000	✓
129	0.0000	✓
145	0.2179	✗

Four of the five original confirmed breakpoints are significant. At regime boundaries, "rules are uncertain"—pulling by multiple sets of rules leads to increased predictive uncertainty.

Quantitative Independence Test of Dimensions (a) and (b). The Spearman rank correlation coefficient between the two dimensions across the 5 confirmed breakpoints with valid (b) data is $\rho = 0.224$ ($p = 0.718$)—extremely low and completely non-significant. This quantitatively supports that the two are distinct statistical fingerprints of regime switching, rather than two facets of the same phenomenon. The contradiction that (a) is significant while (b) is not at $N=145$, together with this extremely low correlation, strongly demonstrates this point.

(c) Random Forest Factor Importance Migration.

Table 7: Random Forest Factor Importance Migration

Confirmed Bkp N	Most Changed Factor	Left Importance	Right Importance	Δ
23	mass_excess_keV	0.546	0.284	−0.262
89	Qa_keV	0.009	0.390	+0.381
108	N_over_Z	0.040	0.185	+0.145
129	mass_excess_keV	0.708	0.499	−0.209
145	mass_excess_keV	0.609	0.464	−0.145

At each confirmed breakpoint, the most important predictive factor undergoes dramatic change. At $N=89$, the importance of Qa_keV surges approximately 40-fold.

3.3 Prediction 2 of the Nuclear Shell Regime-Switching Theory: Cross-Data Subset Replication of Theorem C

Theorem C (Spherical Symmetry Limit Restoration Theorem) has been verified by Tang (2026x)[4] in a specific data environment. This paper replicates this result in a cross-data subset—using the

regime_nuclide_chart.csv merged data (3262 rows), independent of the original data environment of Tang (2026x)[4].

Table 8: Cross-Data Subset Replication of Theorem C ($\sigma = 2.0\text{--}5.0$)

σ	N=20	N=28	N=50	N=82	N=126	All >1.5?
2.0	2.68	3.40	3.73	4.72	6.47	✓
3.0	2.32	2.89	3.18	4.05	5.54	✓
4.0	2.06	2.50	2.76	3.50	4.75	✓
5.0	1.87	2.21	2.43	3.07	4.10	✓

All five traditional magic numbers have enhancement factors greater than 1.5 across all σ values. Enhancement factors decrease monotonically with increasing σ , and increase monotonically from light to heavy nuclei.

Discovery and Correction of the .ssr Trap—An Independent Methodological Contribution. The key to successful replication was the discovery and correction of the .ssr attribute trap in statsmodels' WLS. This attribute returns the weighted residual sum of squares ($\sum w \times \text{resid}^2$), whereas the F-statistic of the Chow test requires the unweighted residual sum of squares ($\sum \text{resid}^2$). The correct approach is to manually compute `np.sum(resid**2)`.

Table 9: Comparison of Enhancement Factors Before and After .ssr Trap Correction ($\sigma = 3.0$)

Magic N	Before Correction (using .ssr)	After Correction (using <code>np.sum(resid²)</code>)
20	0.80	2.32
28	0.87	2.89
50	0.82	3.18
82	0.57	4.05
126	0.74	5.54

Before correction, all enhancement factors were less than 1; after correction, all are greater than 2.0. The successful replication of Theorem C depends on this correction—if the default return of .ssr were used, Theorem C would be erroneously judged as "unreplicable".

4. Discussion

4.1 Repositioning the Shell Correction Method: From "Supplement" to "Compensation"

The statistical fact reported in this paper—shell correction energy extrema adhere tightly to traditional magic numbers (average distance 2.5–3.6 neutrons across three methods) and stay distant from confirmed breakpoints (average distance 11.8–19.6 neutrons, Cohen's $d = 3.23$)—provides a new statistical benchmark for repositioning the physical meaning of shell correction. The randomization test excludes the alternative

explanation of density asymmetry ($p = 0.002$). The characteristic of extrema adhering to traditional magic numbers remains robust to liquid drop model parameter value variation ($\pm 30\%$), grid resolution variation (three window sizes), and functional form variation (adding Wigner terms)—this strongly suggests that this characteristic is a physical fact rather than an artifact of the specific design of the liquid drop model. The random significant breakpoint set test confirms that the confirmed breakpoint network has no significant extra attraction to extrema ($p = 0.060$).

The confirmed breakpoint network does not need shell correction—it is a physical structure more fundamental than shell correction. The success of shell correction precisely demonstrates that independent particle motion requires external compensation—this constitutes independent empirical support for Theorem E of Tang (2026x)[4].

4.2 Physical Significance of Theorem 10

Theorem 10 reveals the existence of a statistical phase transition in the nuclear shell—anomalous features across three dimensions of response function, error structure, and predictive model appear consistently at confirmed breakpoints. All 7/7 pass robustness tests across eight baseline ranges. Singularity remains significant at confirmed breakpoints away from the $N=Z$ line. The Spearman $\rho = 0.224$ between dimensions (a) and (b) quantitatively supports that they are two distinct statistical fingerprints of regime switching.

4.3 Functional Positioning and Complementary Relationship of Information Dualities

The duality of Tang (2026k)[7] establishes the mathematical legitimacy of the overall framework, while the duality of Tang (2026x)[4] establishes the physical interpretability of specific statistical quantities. The two are complementary—the former provides mathematical legitimacy, the latter provides physical interpretability. The singularity of the second-order derivative of binding energy in Theorem 10 of this paper constitutes independent empirical verification of the Tang (2026x) duality in nuclear physics.

4.4 Methodological Significance of the .ssr Trap Correction

The discovery and correction of the .ssr trap has independent methodological value. The successful replication of Theorem C depends on this correction—if the default return of .ssr were used, Theorem C would be erroneously judged as "unreplicable". Other researchers are welcome to replicate Theorem C using independent code bases.

4.5 Testable Predictions and Candid Acknowledgment

Table 10: Summary of Testable Predictions

Prediction	Content	Test Path	Falsification Condition
P1	In future higher-precision nuclear mass measurements, shell correction energy extrema should still adhere tightly to traditional magic numbers (deviation ≤ 5 neutrons)	FAIR/FRIB experimental data + 2D extrema detection of shell correction energy	Extrema deviate from magic numbers by > 5 neutrons
P2	Shell correction energy extrema should not show statistically significant clustering near confirmed breakpoints	Comparison of extrema density near confirmed breakpoints (± 3) vs. near traditional magic numbers	Significant clustering near confirmed breakpoints ($p < 0.05$)
P3	Response function singularity at confirmed breakpoints should reappear in the $N > 150$ region	Local baseline test of second-order derivative of binding energy in superheavy nuclei	No significant peak (ratio < 1.5)
P4	Theorem C can be replicated using independent code bases	Independent researchers replicate using independent code bases	Enhancement $< 1.5\times$ for $\sigma = 2.0\text{--}5.0$

Hierarchical Candid Acknowledgment:

Methodological level: The cross-data subset replication of Theorem C was completed between different code implementations by the same author and does not constitute independent replication in a strict sense.

Argumentation level: The physical interpretation of the "compensation hypothesis" carries a risk of circular definition. The convergent evidence cited in this paper (extrema localization, $\rho = -0.87$, $\eta \approx 0$) is independent in statistical method but all comes from reanalysis of the same set of nuclear mass data. At the current stage, both the "compensation hypothesis" and the "trivial explanation" are reasonable interpretations, and the final judgment distinguishing them comes from future experimental data. The sensitivity analysis of the liquid drop model's functional form (adding Wigner terms) has preliminarily shown that extrema localization is robust to model selection, but more thorough testing requires switching to entirely different liquid drop model frameworks. The construction of random significant breakpoint sets uses the same Chow F statistic as confirmed breakpoints, which may introduce systematic differences in breakpoint position precision.

Data level: Shell correction energy extrema detection has limitations in grid resolution. The intercept/slope ratio includes interpolated and extrapolated values. Missing value sensitivity confirms that Chow F changes by only 10.2% after deleting 10% of data.

Prediction level: P1 and P2 are strongly falsifiable predictions, while P3 and P4 are moderately falsifiable predictions.

5. Conclusions

This paper conducted empirical tests of the core predictions of the nuclear shell regime-switching theory and constructively extended the quantum-classical hybrid framework. The main conclusions include:

(1) First report of the statistical fact that shell correction energy extrema adhere tightly to traditional magic numbers and stay distant from confirmed breakpoints. Three extrema detection methods consistently verify this, the randomization test excludes the alternative explanation of density asymmetry ($p = 0.002$), and extrema localization is robust to liquid drop model parameter value variation, grid resolution variation, and functional form variation. The random significant breakpoint set test confirms that the confirmed breakpoint network has no significant extra attraction to extrema. This statistical fact supports the "compensation hypothesis", but the physical interpretation carries a risk of circular definition, and three testable predictions have been provided for future experimental verification.

(2) First proposal of Theorem 10 (Multidimensional Response Theorem)—multidimensional response function evidence of a statistical phase transition exists at confirmed breakpoints. After local baseline correction, all 7 confirmed breakpoint ratios are significant, and robustness tests across eight baseline ranges pass 7/7. The Spearman $\rho = 0.224$ between dimensions (a) and (b) quantitatively supports that they are two distinct statistical fingerprints of regime switching.

(3) Theorem C was replicated across data subsets ($\sigma = 2.0\text{--}5.0$ all enhanced >1.5 times), and the .ssr attribute trap was first discovered and corrected, providing direct guidance for all researchers using Python for WLS Chow tests.

Data Availability Statement

The Regime Nuclide Map and nuclide regime coordinate database generated in this paper are uploaded to ZENODO along with this paper. Basic nuclear data come from the AME2020 and NUBASE2020 public databases (Wang et al., 2021)[1], comprising six independent data files. The confirmed breakpoint network has been verified across six independent data dimensions. All analysis code is available upon request.

References

- [1] Wang, M., Huang, W. J., Kondev, F. G., Audi, G., & Naimi, S. (2021). The AME 2020 atomic mass evaluation (II). Tables, graphs and references. *Chinese Physics C*, 45(3), 030003. DOI: 10.1088/1674-1137/abddaf
- [2] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. DOI: 10.1023/A:1010933404324
- [3] Chow, G. C. (1960). Tests of equality between sets of coefficients in two linear regressions. *Econometrica*, 28(3), 591–605. DOI: 10.2307/1910133
- [4] Mayer, M. G., & Jensen, J. H. D. (1955). *Elementary Theory of Nuclear Shell Structure*. Wiley.
- [5] Strutinsky, V. M. (1967). Shell effects in nuclear masses and deformation energies. *Nuclear Physics A*, 95(2), 420–442. DOI: 10.1016/0375-9474(67)90510-6

- [6] Tang, S. (2026g). Testability: Factor Hierarchy Theory and Scientific Methodology for Multi-Factor Complex Systems — Complete Empirical Evidence from Global Asset Pricing and a Cross-Disciplinary Application Paradigm. Zenodo. DOI: 10.5281/zenodo.21104307
- [7] Tang, S. (2026k). Information-Theoretic Duality Between Regression Topology and Statistical Mechanics—A Data-Driven Equivalence of the Ehrenfest Classification Validated on the 2D Ising Model as the Physics Cornerstone for the Factor Hierarchy Law. Zenodo. DOI: 10.5281/zenodo.21154577
- [8] Tang, S. (2026m). The Reduction of Universality: A Regime Boundary Theory for Finite-System Phase Transitions — Cross-Model Evidence for Bimodal Splitting, the Precursor Regime Boundary, and the Tool-Dependence Theorem. Zenodo. DOI: 10.5281/zenodo.21258474
- [9] Tang, S. (2026r). From the Law of Meta-Laws to an Operating System for Scientific Discovery: The Axiomatized Core, Philosophical Foundation, and Testability Norms of the Factor Hierarchy Law. Zenodo. DOI: 10.5281/zenodo.21306949
- [10] Tang, S. (2026v). Nuclear Shell Regime Audit: The Magic-Number Shift Theorem and the Discovery of the N=14 Statistical Rupture—The η Hierarchy Structure, Bai-Perron Optimal 10 Breakpoints from Zero-Preset Exhaustive Search, and Six Quantitative Theorems. Zenodo. DOI: 10.5281/zenodo.21621800
- [11] Tang, S. (2026w). Regime-Switch Network and the True Structure of the Nuclear Shell—Confirmed Breakpoint Networks, Transition Zone Width Theorem, and the Fifteen New Discoveries. Zenodo. DOI: 10.5281/zenodo.21622347
- [12] Tang, S. (2026x). The Triple Demotion of the Nuclear Shell: The Shell Model, the Liquid Drop Model, and the Independent-Particle-Motion Axiom—Based on 2D Ising Calibration and Continuous Sphericity Weight Restoration. Zenodo. DOI: 10.5281/zenodo.21634385
- [13] Tang, S. (2026y). Theoretical Extensions and Methodological Outputs of the Quantum-Classical Hybrid — Regime Factor Purity Spectrum, Confirmed Breakpoint Network Binary Structure, and Cross-Disciplinary Predictions. Zenodo. DOI: 10.5281/zenodo.21635378
- [14] Tang, S. (2026z). Technical Specification of the Scientific Discovery Operating System: Mathematical Formalization and Engineering Architecture—Based on the Factor Hierarchy Law and Testability Norms. Zenodo. DOI: 10.5281/zenodo.21638618
- [15] Tang, S. (2026aa). The Regime Nuclide Map (I): Cross-Property Verification, Two-Dimensional Structure, and Network Update of the Confirmed Breakpoint Network——Predictive Test and New Discoveries Based on the Nuclear Shell Regime-Switching Theory. Zenodo. DOI: 10.5281/zenodo.21896164